

Migrating SQL Server to Azure

Exhaustive inventory of targets, methods & tools | v1.1

July 2026

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Goal. Exhaustively list every way and every tool to migrate a SQL Server	

database to an Azure service (all PaaS, including containers) or a VM (Azure VM / Azure VMware Solution).

Audience. Partners, architects, and customer DBAs — usable in pre-sales and as the knowledge base behind the *SQL in a Day* AI Migration Agent (§14).

Verification. Tool retirements, version requirements and target families were cross-checked against Microsoft Learn and product announcements (current as of July 2026). Links are gathered in §16 Sources.

Version. v1.1 — July 2026. Change history in §17 Document version & changelog.

IMPORTANT 2025-2026 tooling reset — read this first. - **Data Migration Assistant (DMA)** is RETIRED (16 July 2025). - **Azure Data Studio (ADS)** and its Azure SQL Migration extension are RETIRED (28 February 2026). - The new entry point is the **SSMS 22 Migration Component** (assess + migrate from SSMS), complemented by SQL Server migration in Azure Arc (portal, with Copilot) and Azure Database Migration Service (DMS). - Modern **DMS** supports offline-only migration to Azure SQL Database (online / minimal-downtime is available for Managed Instance and SQL VM targets, per the [DMS supported-scenarios](#) matrix). - **Azure DMS classic** SQL Server scenarios are RETIRED (15 March 2026) — use the modern DMS (an Azure resource, via portal / PowerShell / CLI).

1 Why migrate in 2026 (the short “why now”)

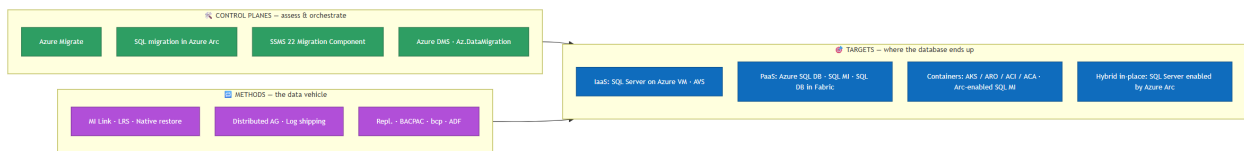
- **AI runs on data.** Modern, managed databases are the foundation; SQL Server 2025 / Azure SQL add a native vector type + functions, **DiskANN** vector indexing (high-throughput ANN search), native json, and zero-ETL **Fabric Mirroring** (analytics on OneLake with no pipelines) — making the estate AI-ready and improving migration (enhanced distributed AG).
- **End-of-support pressure.** Out-of-support SQL Server/Windows versions push modernization; Extended Security Updates (ESU) are free on Azure VMs, and for on-prem are delivered only via Azure Arc (paid) — this changes the *stay vs migrate* math.
- **Cost levers.** Azure Hybrid Benefit (AHB) applies to SQL DB (vCore), Managed Instance and VM (not to Fabric SQL DB) — up to ~85% savings when combined with reservations/ESU.
- **Partner leverage.** Sharing a deal with a partner increases win rate and deal size; programs like Cloud Accelerate Factory and SQL in a Day industrialize delivery (§14). Commercial & funding levers are detailed in §15.
- **Concrete forcing function.** SQL Server 2016 reaches End of Support on 14 July 2026 — the #1 trigger in EMEA. “Stay” then means paying ESU; “migrate” makes ESU free on Azure.

Version	Extended End of Support	Status (Jul 2026)
SQL Server 2012	12 Jul 2022 — ESU ended Jul 2025	✗ Out of support
SQL Server 2014	9 Jul 2024 — ESU ~Jul 2027	Out of support (ESU window)
SQL Server 2016	14 Jul 2026 — ESU ~Jul 2029	EOS imminent
SQL Server 2017	12 Oct 2027	✓ Extended support
SQL Server 2019	8 Jan 2030	✓
SQL Server 2022	11 Jan 2033	✓
SQL Server 2025	GA 18 Nov 2025	✓ Latest

Windows Server often sits under SQL Server — **WS 2012/2012 R2** is out of support (ESU to Oct 2026), WS 2016 EOS ~Jan 2027. Factor the OS into the move.

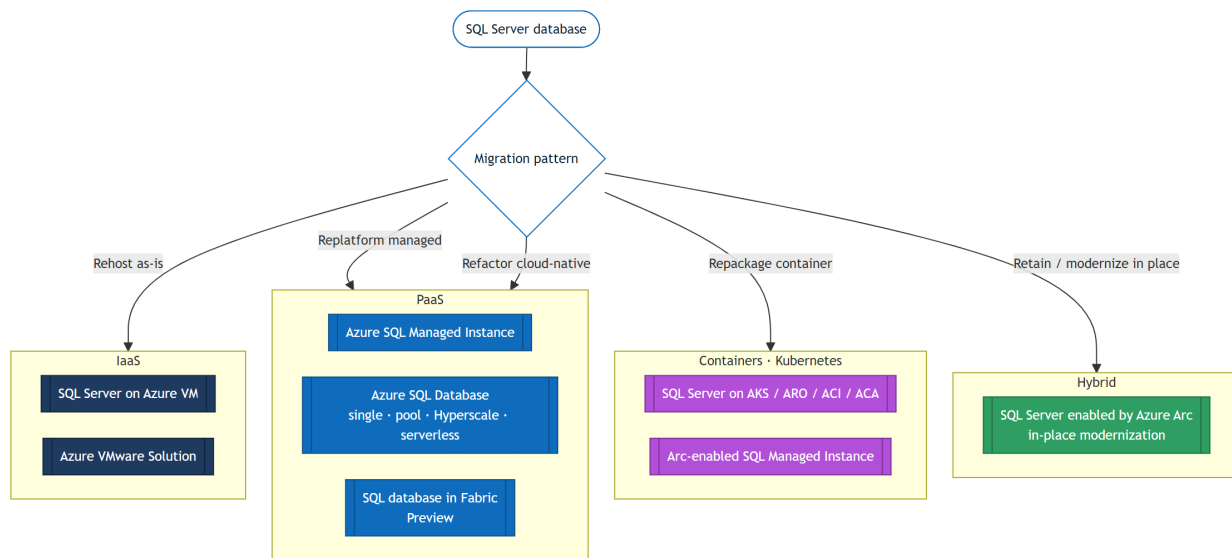
2 Taxonomy — separate targets, control planes, and methods

The single most common mistake is to mix these three layers. Keep them distinct:



- **Targets** = runtime destinations (where the DB lives).
- **Control planes / experiences** = how you assess, recommend and orchestrate (Azure Migrate, Arc, SSMS 22, DMS). Azure Arc-enabled SQL Server is a control plane, not a runtime target.
- **Methods** = the actual data-movement vehicle (MI Link, LRS, backup/restore, replication...).

3 The Azure targets (8 families)



#	Target	Layer	When to choose it	Compatibility	Doc
1	SQL Server on Azure VM	IaaS	Faithful lift & shift: OS / file-system control, exact version, FileStream/FileTable, PolyBase, cross-instance DTC, third-party agents.	Full	overview
2	Azure VMware Solution (AVS)	IaaS	Zero-refactor data-center exit for existing VMware estates; keeps FCI and Always On AG; migrate with VMware HCX / vMotion.	Full	AVS
3	Azure SQL Managed Instance	PaaS	Managed lift-and-shift: keep instance objects (logins, SQL Agent, server triggers, cross-DB, linked servers), native vNet. Tiers GP / BC / Next-gen GP (GA since Nov 2025 — Elastic SAN backend: 500 DBs, 128 vCores, 32 TB, 80K IOPS, 192 MB/s log, 3–4 ms latency, configurable IOPS/memory; ~5x better price-per-DB by density). Free offer: 1 instance / 12 months. No Hyperscale on MI.	~Near-full (instance)	overview
4	Azure SQL Database	PaaS	Cloud-native app / microservice. Models: single DB / elastic pool; tiers GP / BC / Hyperscale; purchasing vCore / DTU / serverless. Hyperscale scales to 128 TB (large / HTAP); serverless for intermittent; elastic pools for consolidation. Free offer: 10 serverless DBs for the subscription lifetime.	Database surface (no instance-level)	overview
5	SQL database in Fabric (Preview for migration)	PaaS	Fabric-native OLTP unified with OneLake. Migrate via Fabric Migration Assistant (DACPAC schema ≤ 20 MB, on-prem data gateway only, no Private Link). Not an enterprise OLTP target yet.	Subset; preview	Migration Assistant
6	SQL Server in containers — AKS / ARO / ACI / ACA	Container	Full control of the engine in a container (dev/test, edge, custom). Pod + PersistentVolume; HA via the Kubernetes scheduler.	High — SQL on Linux (no FILESTREAM/FileTable, SSRS/SSAS/SSIS, ML Services; SQL Agent off by default)	SQL on Kubernetes
7	Azure Arc-enabled SQL Managed Instance	Container (PaaS)	Managed SQL MI engine on any Kubernetes (AKS, ARO, EKS, GKE, OpenShift) via kubectl + CRD. Sovereignty / edge / multi-cloud.	~Same as SQL MI	create Arc SQL MI
8	SQL Server enabled by Azure Arc	Hybrid (control plane)	Not a runtime target — the bootstrap pillar: connect on-prem SQL <i>without touching it</i> → weekly continuous assessment, blocker detection, ESU, PAYG (OpEx) licensing, governance, and a portal Copilot-assisted Database migration to MI/VM (MI Link / LRS). Sources SQL 2012+.	n/a	Arc migration

Lift & shift is not VM-only. Both SQL Server on Azure VM (IaaS) and Azure SQL Managed Instance (managed PaaS) are valid lift-and-shift targets. Since 2024, MI compatibility (SQL Agent, cross-DB queries, linked servers...) makes it the default *managed* lift-and-shift.

4 Control planes & assessment experiences

Tool / experience	Role	Status (2026)	Notes
Azure Migrate	Discovery / assessment / sizing / business case at scale	GA (+ Arc-based agentless discovery, Preview)	Appliance (VMware/Hyper-V/Physical) or import-based or Arc-based. Right-sizes SQL DB / MI / VM.
SQL Server migration in Azure Arc	Portal-driven assess + migrate for any Arc-enabled SQL Server	GA; MI target GA, VM target Preview	Copilot recommends method (MI Link vs LRS); continuous assessment; supports sources from SQL Server 2012+.
SSMS 22 Migration Component	DBA-first entry point: assess + launch the right path from SSMS	GA (Windows-only)	Replaces DMA / ADS extension. Backup/restore, MI Link, DMS. Azure SQL assessment capability expected ~Q3 CY2026 (<i>public roadmap, subject to change</i>).
Azure DMS (modern)	Managed migration orchestration (Azure resource · portal / PowerShell / CLI)	GA	Use the modern DMS — DMS <i>classic</i> SQL scenarios retired 15 Mar 2026. Offline-only to Azure SQL DB; online/minimal-downtime to MI / SQL VM (MI Link preferred for MI).
PowerShell Az.DataMigration / Azure CLI	Automate DMS at scale (CI/CD)	GA	Often the only viable path beyond ~50 databases.
SSMA	Heterogeneous conversion (schema/code/data)	GA	Oracle / Sybase / DB2 / MySQL / Access → Azure SQL. Not for homogeneous SQL→SQL.
Database Experimentation Assistant (DEA)	Capture + replay a production workload on the target to validate performance <i>before</i> cutover	Retired 15 Dec 2024	Replaced by the open-source SQLWorkload / WorkloadTools. The <i>concept</i> (catch plan/compat regressions before cutover) remains valuable; the MS tool itself is retired.

NOTE Retired — do not use in new runbooks: DMA (16 Jul 2025) and Azure Data Studio + Azure SQL Migration extension (28 Feb 2026). For cross-platform SQL dev, use VS Code + MSSQL extension; migration work continues via SSMS 22 / Azure Arc / DMS / [Az.DataMigration](#) CLI. SQL Data Sync retires 30 Sep 2027 — don't build new sync/migration on it (use ADF, transactional replication or AG).

4.1 Microsoft tooling by source → target (assess · data · schema)

The Microsoft-recommended tool per source/target combination (homogeneous and heterogeneous):

Source	Target	Assess	Data migration	Schema
SQL Server (Arc-enabled)	Azure SQL MI	SQL migration in Azure Arc	SQL migration in Azure Arc	Not needed
SQL Server (not Arc)	Azure SQL VM / MI	Azure Migrate	DMS	Not needed
SQL Server	Azure SQL DB	Azure Migrate	STRIM (online) · DMS (offline)	DMS
Sybase	Azure SQL	Azure Migrate	STRIM	SSMA for Sybase
Oracle	Azure SQL	Azure Migrate	STRIM	SSMA for Oracle

GitHub Copilot is applicable as AI-assisted migration tooling (schema/code conversion inside SSMA; app-level via GitHub Copilot App Modernization for .NET / Java). Portfolio & application/code assessment partners complementary to Azure Migrate: Dr Migrate, CAST Highlight, UnifyCloud.

5 Migration methods per target

Standardized columns (Microsoft Learn style): **Method · Min source · Target/min · Downtime · Key constraints.**

5.1 To SQL Server on Azure VM (IaaS rehost)

Method	Min source	Downtime	Key constraints / notes
Azure Migrate (lift & shift)	SQL 2008 SP4	Online (replication)	Whole VM/instance, incl. FCI and AG; up to ~35,000 VMs.
Distributed availability group (DAG)	SQL 2016	Near-zero	Reuse on-prem AG; needs AD Domain Services (or workgroup AG + certs) and ports open.
Backup to a file (.bak) + copy	SQL 2008 SP4	Offline	Simple, supports > 1 TB; use compression / multi-file split for WAN.
Backup to URL (Azure Blob)	SQL 2014	Offline	12.8 TB (SQL 2016+) / 1 TB otherwise; for > 1 TB use local backup + AzCopy.
Detach & attach (MDF/LDF via Blob)	SQL 2008	Offline	For very large DBs where backup/restore is too slow.
Log shipping	SQL 2008	Minimal	Windows-only (not for SQL on Linux sources).
Always On AG	SQL 2012	Near-zero	Fail an existing AG onto Azure VM replicas.
Convert machine to VHD / Ship hard drive / Azure Data Box	any	Offline	Estate exit with limited WAN; multi-TB .bak / .bacpac via Data Box.

5.2 To Azure SQL Managed Instance (managed lift-and-shift)

Method	Min source	Downtime	Key constraints / notes
Managed Instance link (MI Link)	SQL 2016 and later (incl. 2022, 2025; Win Server 2016+, Ent/Std/Dev)	Near-zero (online)	Distributed-AG based; R/O readable target during migration; reverse fallback to SQL 2022 / 2025 (DR / Azure exit); up to 10 simultaneous DBs (Arc ext 1.1.3348.364+); needs port 5022 both ways.
Log Replay Service (LRS)	SQL 2012 (all editions)	Offline (planned)	Full/diff/log → Azure Blob; public endpoint; 30-day max window; FULL recovery + NORECOVERY (no reads); the only managed path that applies differentials to MI; GP & BC supported, but for large Business Critical migrations prefer MI Link (true online).
Native backup & restore (.bak)	SQL 2008	Offline	Simplest; migrate TDE certificate <i>before</i> restore or it fails late; master/msdb restore not supported (script instance objects).
Transactional replication	SQL 2012	Online	Replicate all/part; article-type limits.
bcp / Smart Bulk Copy	any	Offline	High-speed data-only / partial (parallel copy).
BACPAC / SqlPackage	any	Offline	Smaller DBs / simple.
Azure Data Factory — Copy	any	Offline / batch	When migration = integration / transformation.

5.3 To Azure SQL Database (cloud-native refactor)

Method	Min source	Downtime	Key constraints / notes
Azure DMS (offline)	SQL 2008+	Offline	Offline only to Azure SQL DB (online/minimal-downtime is available for MI / SQL VM, not SQL DB).
STRIM (online CDC)	any	Online (near-zero)	Microsoft-recommended online / CDC data migration to Azure SQL DB — fills the gap that DMS is offline-only for SQL DB; pair with DMS for schema.
Transactional replication	SQL 2016–2019 only	Online	Push subscription only; subset of tables/columns/rows; article-type limits (no <code>hierarchyid</code> , <code>sql_variant</code> ...).
BACPAC / SqlPackage	any	Offline	Small/medium; SqlPackage for scale.
bcp / Smart Bulk Copy	any	Offline	Data-only / bulk.
Azure Data Factory — Copy	any	Offline / batch	BI / integration.

X Not supported to Azure SQL Database: native .bak restore, detach/attach, MI Link. SQL Agent → use [Elastic Jobs](#).

DACPAC vs BACPAC. A DACPAC packages the schema only; a BACPAC packages schema + data. Both are produced/consumed by SqlPackage and the

VS Code MSSQL extension — a simple way to migrate (BACPAC) or version (DACPAC) offline.

5.4 To SQL database in Fabric (*Preview*)

Method	Downtime	Key constraints / notes
Fabric Migration Assistant — DACPAC	Offline	Schema via DACPAC ≤ 20 MB; AI-assisted compatibility fixes; data via Fabric Data Factory copy job + on-prem data gateway (no VNet gateway / Private Link).
Fabric Mirroring for SQL Server	n/a (continuous)	NOT a one-shot migration — near-real-time CDC replication to OneLake for analytics. GA (Nov 2025), optimized for SQL Server 2025. Complementary "analytical modernization" path.

5.5 To Containers / Arc-enabled SQL MI

Target	Method	Downtime	Notes
Arc-enabled SQL MI (AKS/ARO/...)	Native backup/restore, point-in-time restore	Offline	Requires Arc data controller; exposes a SQL MI endpoint → logical vehicles of \$5.2 apply.
SQL Server container (mcr image)	Backup/Restore via mounted volume, detach/attach, BACPAC, bcp, ADF	Offline / Online (repl.)	Persist on Azure Disk (AKS/ARO) / Azure Files (ACI/ACA); HA via scheduler.

WARNING Containers are not a fully-managed cloud database. Plain mssql containers (AKS / ARO / ACI / ACA) put patching, high availability and backups entirely on you — best suited to dev/test and edge scenarios. Arc-enabled SQL MI automates engine patching, backups and HA through the Arc data controller, but you still own the Kubernetes cluster, persistent storage and DR. Neither is equivalent to the fully-managed Azure SQL Database / Managed Instance PaaS.

5.6 Offline & network transfer accelerators (large estates)

For multi-terabyte databases the network — not the method — is usually the bottleneck. Plan the **bulk transfer / initial seed** explicitly:

Accelerator	Use it for	Notes
Azure Data Box / Data Box Heavy	One-shot physical move of multi-TB .bak / .bacpac or a whole VM fleet	Data Box ≈ 80 TB usable; Data Box Heavy ≈ 770 TB for hundreds of TB. Microsoft recommends Data Box to migrate a SQL server/estate and as the initial <i>seed</i> before syncing the delta over the network.
ExpressRoute / dedicated circuit	Sustained high-throughput online transfer & seed	Avoids WAN time-outs on long-running backup/restore, LRS or replication.
AzCopy + Backup-to-URL	Push large local backups to Blob	For DBs > 1 TB: local backup + AzCopy is faster/safer than direct Backup-to-URL.
Azure Storage Mover	Orchestrated bulk file transfer on-prem → Azure Files / Blob	Centralized, resumable; good for backup repositories.
Compressed / split / encrypted backups	Shrink & secure the payload in transit	Compression + multi-file split optimize WAN; backup encryption (SQL 2014+) for compliance during transit.

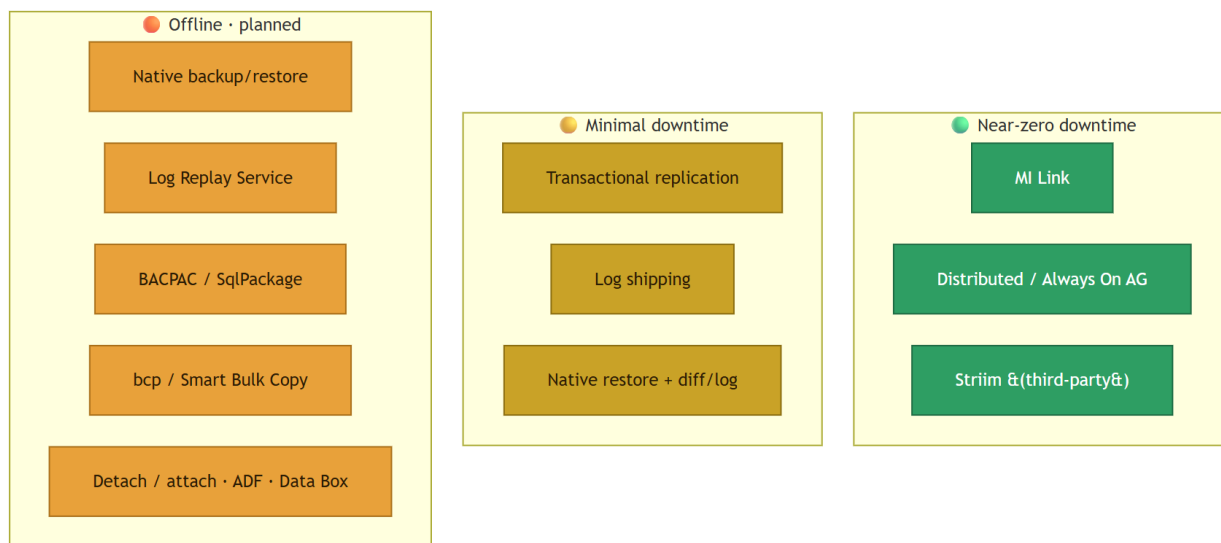
Seed-then-sync pattern. For very large or low-downtime migrations: ship the initial full backup via Data Box (or AzCopy over ExpressRoute), then catch up the delta with log restore (LRS), MI Link, transactional replication or log shipping before cutover.

6 Ancillary components (the cutover blockers)

Databases rarely move alone — these sub-components block go-live if forgotten:

Component	Path to Azure
SQL Agent jobs	MI: native · SQL DB: Elastic Jobs
Logins / users	Script + recreate; DMS does not migrate Windows logins by default (enable option + grant MI read access to Entra ID).
SSIS	Azure-SSIS Integration Runtime (SSISDB via DMS).
SSRS	Power BI paginated reports (RDL).
SSAS	Azure Analysis Services or Power BI Premium (XMLA).
Linked servers / cross-DB	Supported on VM/MI; not on SQL DB — refactor required.
TDE	Migrate the server-level certificate <i>before</i> any native restore to MI.

7 Downtime strategy (cutover window) — the #1 architect criterion



Sizing rule. Never size MI/SQL DB on *average* CPU. Use a Perfmon baseline ≥ 7 days + ~20% headroom for cutover. Network: don't estimate size ÷ bandwidth — Backup-to-URL throughput is capped by the Blob layer; test with AzCopy + one full backup before committing a go-live date, and plan a rollback / fallback window in case the estimate proves optimistic.

8 Summary matrix — method / tool × target

Method / tool	SQL VM	AVS	SQL MI	SQL DB	Fabric SQL DB	Arc SQL MI	SQL container
Azure Migrate (assess)	✓	✓	✓	✓	–	–	–
DMS	✓	–	✓ (online)	✓ (offline)	✗	–	–
MI Link	reverse	✗	✓	✗	✗	✓	✗
Log Replay Service	✗	✗	✓	✗	✗	–	✗
Native backup/restore	✓	✓	✓	✗	✗	✓	✓
Distributed / Always On AG	✓	✓	✗	✗	✗	✗	✗
HCX / vMotion	✗	✓	✗	✗	✗	✗	✗
Transactional replication	✓	✓	✓	✓ ¹	✗	✓	✓
BACPAC / SqlPackage	✓	✓	✓	✓	✓ (DACPAC)	✓	✓
bcv / Smart Bulk Copy	✓	✓	✓	✓	–	✓	✓
ADF Copy	✓	✓	✓	✓	✓	✓	✓
Fabric Migration Assistant	✗	✗	✗	✗	✓	✗	✗
SSMA (heterogeneous)	✓	✓	✓	✓	–	✓	✓

✓ supported · ✗ n/a · – indirect/non-first-class · reverse only · ¹ SQL DB transactional replication: source **SQL 2016–2019** only.

9 Source-version & retirement reference

Tooling timeline: DMA retired Jul 2025 → ADS + SQL Migration extension retired Feb 2026 → DMS *classic* SQL scenarios retired Mar 2026 → SQL Data Sync retires Sep 2027.

Item	Status / requirement
DMA	Retired 16 Jul 2025 → SSMS 22 / Arc / Azure Migrate
Azure Data Studio + SQL Migration extension	Retired 28 Feb 2026 → VS Code + MSSQL; SSMS 22 / DMS
DMS <i>classic</i> — SQL Server scenarios	Retired 15 Mar 2026 → modern DMS (portal / PowerShell / CLI)
Database Experimentation Assistant (DEA)	Retired 15 Dec 2024 → open-source SQLWorkload / WorkloadTools
SQL Data Sync	Retires 30 Sep 2027 → use ADF / transactional replication / AG
DMS → Azure SQL DB	Offline only (online available for MI / SQL VM)
MI Link source	SQL Server 2016 and later (incl. 2022, 2025), Windows Server 2016+, Ent/Std/Dev
LRS source	SQL Server 2012+, all editions
Native restore → MI	SQL Server 2008+
Transactional replication → MI	SQL Server 2012+
Transactional replication → SQL DB	SQL Server 2016–2019 only
Backup-to-URL size	12.8 TB (2016+) / 1 TB
Fabric Migration Assistant	Preview; DACPAC ≤ 20 MB; on-prem gateway only
SQL Server 2025 (source)	GA 18 Nov 2025 — native vector/JSON, improved DAG

10 Cross-cloud sources & reverse migration

- **Cross-cloud sources** explicitly supported by DMS / LRS / MI Link / native restore: AWS EC2, AWS RDS for SQL Server, GCP Compute Engine, GCP Cloud SQL for SQL Server.
- **Reverse / exit migration:** SQL DB → SQL Server via BACPAC + scripts (heavy); MI → SQL Server 2022 / 2025 via MI Link reverse failback (trivial — a strategic portability argument).
- **Neutral non-Microsoft targets** (for honest comparison): AWS RDS for SQL Server (+ AWS DMS), GCP Cloud SQL (+ GCP DMS), Tessell, OCI self-managed. Azure differentiators: MI Link, Arc, Fabric Mirroring.

11 Third-party alternatives (when they beat the native stack)

Tool	Better when...
Striim	Microsoft-recommended online / CDC data-migration vehicle for SQL Server → Azure SQL Database (the online path DMS lacks for SQL DB) and for heterogeneous sources (Sybase / Oracle / DB2 → Azure SQL, paired with SSMA for schema); also real-time CDC to Event Hubs / Synapse / Cosmos in parallel.
Dr Migrate / CAST Highlight / UnifyCloud	Portfolio & application/code assessment at scale (wave planning, dependency mapping) — complementary to Azure Migrate.
Qlik Replicate	Heterogeneous sources (Oracle, DB2, iSeries, SAP HANA) with in-flight transforms — more mature than DMS for Oracle→SQL.
Fivetran HVR	Broad CDC + observability; multi-target (Snowflake + Fabric).
Quest SharePlex	SQL Server ↔ Oracle replication; de-Oracle-ization projects.
Carbonite Migrate	VM-level always-on with failback when Azure Migrate/ASR can't handle the OS/hypervisor (KVM, Xen, Citrix).
Veeam / Commvault / Cohesity / Rubrik	Already in place — push backups to Blob, restore on SQL VM (no double licensing).
Redgate / Liquibase / Flyway	Schema versioning (GitOps) post-migration — SqlPackage's weak spot.
VMware HCX (Broadcom)	The zero-downtime "as-is" option for complex VMware clusters incl. SQL FCI → AVS.

12 Decision criteria & “when to recommend what”

Criterion	VM	AVS	MI	SQL DB	Fabric SQL DB	AKS / Arc
OS access required	✓	✓	✗	✗	✗	partial
Cross-DB / DTC transactions	✓	✓	✓	✗	✗	✓
FILESTREAM / FileTable	✓	✓	✗	✗	✗	✗
PolyBase	✓	✓	✗	✗	✗	✓
Service Broker	✓	✓	✓ ¹	✗	✗	✓
SQL CLR / linked servers	✓	✓	✓	✗	✗	✓
SQL Agent	✓	✓	✓	✗ (Elastic Jobs)	✗	✓
Min. downtime achievable	~h (DAG)	~h (vMotion)	~min (MI Link)	min-h	h	depends
Managed patch / upgrade	Auto-patch	✗	✓ Evergreen	✓ Evergreen	✓ Evergreen	✗
Azure Hybrid Benefit	✓	✓	✓	✓ (vCore)	n/a	✓
Sovereignty / edge	✓	✓	limited regions	limited regions	limited regions	✓ (Arc)

¹ Azure SQL MI supports Service Broker **within a single instance** only — no cross-instance Service Broker routing.

Client profile	Neutral recommendation	Why
Banking / regulated / strong on-prem	SQL VM + ESU via Arc, or AVS	OS control, compliance, soft transition
Multi-tenant SaaS ISV	SQL DB Hyperscale / Elastic Pool	Per-tenant elasticity, cost control
Heavy legacy ERP (SAP, on-prem Dynamics)	SQL MI or SQL VM	Instance & SQL Agent compatibility
Modern micro-services app	SQL DB serverless	Pay-per-use, auto-pause
Fabric-native / analytics-first	SQL DB in Fabric + Mirroring	OLTP + OneLake unification
Edge / sovereign / multi-cloud	Arc-enabled SQL MI on local AKS	Sovereignty + Azure consistency
Short-term data-center exit	AVS + VMware HCX	Zero refactor, vMotion
Modernize off end-of-support	SQL VM “as-is” + free ESU	3 years free ESU on Azure VM

13 Field insights — recurring pitfalls

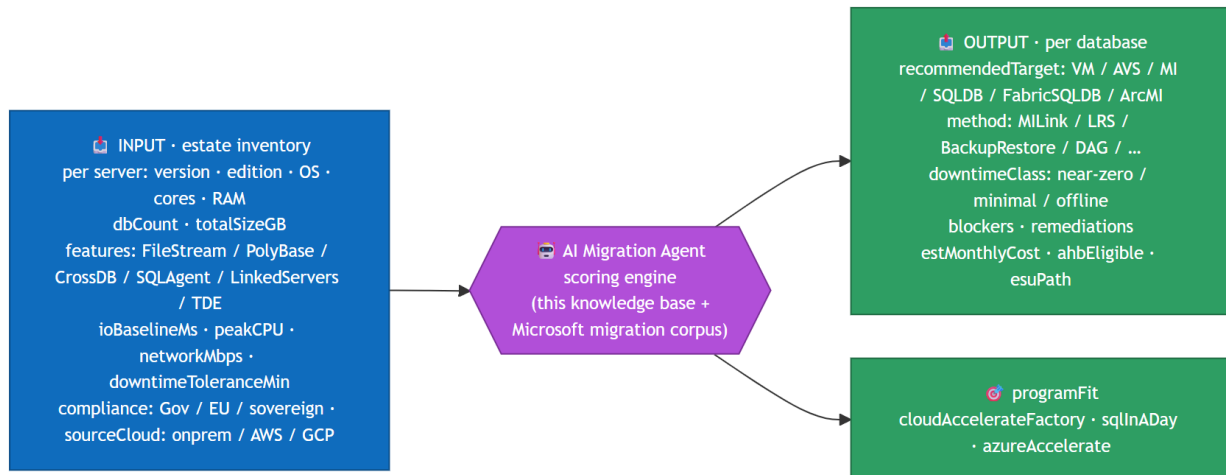
- **TDE certificate:** protect-then-restore order matters — migrate the server cert first, or native restore fails ~80% in with no clear message.
- **Windows logins:** DMS skips them by default; enable the option and grant MI read to Entra ID (Privileged Role Administrator).
- **MI Link ports:** needs 5022 both directions — frequently blocked in banking/industrial networks without a security exception.
- **Other methods need network too:** DMS / LRS / transactional replication require outbound HTTPS (443) to Azure Storage/Blob, SQL 1433 (and 1434/UDP SQL Browser for named instances) — anticipate firewall/NSG blocks in locked-down environments.
- **DAG:** requires AD Domain Services (or workgroup AG + certs) — an infra blocker architects forget.
- **Transactional replication → SQL DB:** SQL 2016–2019 publishers only, with article-type limits — audit before promising.
- **Hyperscale:** the only viable SQL DB choice above 4 TB or with heavy concurrent write I/O.

- **Fabric SQL DB:** Preview for migration, 20 MB DACPAC cap, no Private Link — don't position as enterprise prod yet.
- **Mirroring ≠ migration:** it's continuous analytics replication; treating it as a one-shot migration is a dangerous shortcut.
- **Dependency mapping:** ~60% of migrations stumble on undocumented linked servers and SQL Agent jobs — run a dependency map (Azure Migrate or third-party) before committing a target.
- **SLAs differ:** MI BC 99.99% · SQL DB BC 99.995% (zone-redundant) · SQL DB Hyperscale 99.99% · SQL VM depends on the AG. Align to the app, not a slogan.
- **Security by design:** SQL DB has a public endpoint by default — recommend Private Endpoint + Entra-only auth + disabled SQL auth from day one.
- **Land the zone first:** stand up an Azure **landing zone** (CAF — IAM, policy, networking, monitoring, Defender pre-baked) before migrating — Microsoft reports ~4x faster migration/modernization once it's deployed, because each workload move stops being a one-off.
- **Fabric SQL DB ≠ Azure SQL DB:** not a drop-in replacement — fine-grained security (RLS / OLS / dynamic masking) does not propagate to OneLake; governance must be re-implemented in Fabric. For analytics on operational data, prefer Mirroring (keep OLTP on Azure SQL, mirror into Fabric, no ETL) over a hard cutover.
- **Bootstrap via Arc first:** connect the estate to Azure Arc *before* migrating — free weekly continuous assessment, blocker detection, ESU and PAYG (OpEx) licensing while you plan, with zero workload change (§3 row 8).

14 FY27 SQL Motion context & AI Migration Agent

This document is the knowledge base behind the **FY27 EMEA EPS — Data Motion “SQL in a Day”**. Two notes:

- **Storyline correction.** The deck line “*Azure Migrate, DMA, DMS, SSMA, Cloud Accelerate Factory*” must drop DMA (retired) → use “*Azure Migrate, SSMS 22 + Arc-based assessment, DMS, SSMA, Cloud Accelerate Factory*”. A 4th target pillar — SQL Server enabled by Azure Arc — should appear alongside VM / MI / SQL DB.
- **AI Migration Agent — I/O contract.** The afternoon agent scores the best path on the customer estate. Suggested flow so it's usable by both humans and automation:



Microsoft programs to attach: Cloud Accelerate Factory (zero-cost delivery), Azure Accelerate / FastTrack, AHB + ESU via Arc — detailed in §15.

15 Commercial levers & funding programs (FY27)

Two money levers to combine on every deal: **(A) Microsoft-funded engagement** money (assess / pilot / migrate) and **(B) commercial / licensing** levers that durably cut TCO. **Amounts, ratios and program names are partner-confidential and drift each fiscal year (FY27 starts 1 Jul 2026) — re-validate live in Partner Center / the MCI guide before quoting.**

15.1 Commercial / licensing levers (permanent TCO reducers)

Lever	What it saves	Mechanics
Azure Hybrid Benefit (AHB)	30%+ on Azure SQL DB / MI; compute-only on VM	License + Software Assurance reallocated to Azure; 1 Enterprise core = 4 GP vCores; 180-day dual-use during migration; portal toggle. Not for the DTU model or serverless.
Free ESU on Azure	Removes on-prem ESU cost	Azure VMs / AVS: ESU free & automatic; Azure SQL PaaS: ESU charges stop; on-prem / multicloud via Arc: ESU paid; non-prod free when prod runs ESU via Arc.
PAYG licensing via Arc	Turns the SQL license into OpEx	Billed only when SQL runs; CALs included; can sit inside a MACC. Requires active SA or PAYG enabled.
Free Azure SQL offers	Zero-cost POC / pilots	MI free 12 months; SQL DB free for the subscription lifetime (serverless GP).
Reservations / Savings Plans	1- / 3-yr commitment discount	Stacks with AHB. Partner Earned Credit (15%) does not apply to reservations.
Savings plan for databases	up to 35% on Azure SQL (DB / MI)	1- or 3-year hourly compute commitment; auto-applies across participating database services up to the commitment; stacks with AHB.

15.2 Microsoft-funded engagement programs

Program	Funds	Access
Azure Accelerate (FY26 umbrella = <i>Azure Migrate & Modernize</i> + <i>Azure Innovate</i> + <i>Cloud Accelerate Factory</i>)	Assessments, POC, pilots, deployments, Azure credits, skilling	Self-serve — Partner Center nomination → POE → paid ≤ 45 days
Azure Accelerate for Databases	SQL / data-estate modernization	Same workflow (database scenario)
Cloud Accelerate Factory	Zero-cost Microsoft expert deployment	Via Azure Accelerate in Partner Center
FastTrack for Azure	Free Microsoft engineering guidance (not cash)	Account team / program alias
ECIF (End-Customer Investment Funds)	Assessments, POCs, migrations, training	Field-led via PDM / AE — not self-serve
Azure Frontier Offer (ECIF + ACO, dual-run offset)	Services + dual-run during migration; competitive Oracle / legacy displacement	Field-led via PDM; typically ≥ 50% ACR from Fabric / first-party Databases / AI

15.3 Partner mechanics

- **Attribution first** (funding prerequisite): link the customer via PAL (Partner Admin Link) or DPOR → unlocks the 15% Partner Earned Credit on managed Azure consumption.
- **Routes:** *Self-serve* (Partner Center) — Azure Accelerate, Databases, Cloud Accelerate Factory · *Field-led* (engage the PDM early) — ECIF, Azure Frontier Offer · *Customer self-serve* (Azure portal) — AHB, ESU, free offers, reservations.
- **Typical sequence:** Azure Migrate discovery + TCO → nominate assess/pilot in Azure Accelerate (Databases) → if competitive or ≥ 50% data ACR, engage PDM for the Azure Frontier Offer → pilot on free SQL DB / MI → CAF for zero-cost deploy → POE → lock run-rate with reservations.

16 Sources (Microsoft Learn)

Overviews & taxonomy - Database Migration hub — <https://learn.microsoft.com/en-us/data-migration/> - SQL Server → Azure SQL Database — <https://learn.microsoft.com/en-us/data-migration/sql-server/database/overview> - SQL Server → Azure SQL Managed Instance — <https://learn.microsoft.com/en-us/data-migration/sql-server/managed-instance/overview> - SQL Server → SQL Server on Azure VM — <https://learn.microsoft.com/en-us/data-migration/sql-server/virtual-machines/overview> - Azure SQL feature comparison — <https://learn.microsoft.com/en-us/azure/azure-sql/database/features-comparison>

Control planes / tools - Azure Migrate (SQL assessment) — <https://learn.microsoft.com/en-us/azure/migrate/how-to-create-azure-sql-assessment> - SQL Server migration in Azure Arc — <https://learn.microsoft.com/en-us/sql/sql-server/azure-arc/migration-overview> - SSMS (download/overview) — <https://learn.microsoft.com/en-us/sql/ssms/sql-server-management-studio-ssms> - Azure Database Migration Service — <https://learn.microsoft.com/en-us/azure/dms/dms-overview> - DMS supported scenarios (offline/online per target) — <https://learn.microsoft.com/en-us/azure/dms/resource-scenario-status> - Az.DataMigration PowerShell — <https://learn.microsoft.com/en-us/powershell/module/az.datamigration/> - SSMA — <https://learn.microsoft.com/en-us/sql/ssma/sql-server-migration-assistant> - Database Experimentation Assistant (DEA) — <https://learn.microsoft.com/en-us/previous-versions/sql/dea/database->

[experimentation-assistant-overview](https://learn.microsoft.com/en-us/sql/tools/whats-happening-azure-data-studio) - What's happening with Azure Data Studio — <https://learn.microsoft.com/en-us/sql/tools/whats-happening-azure-data-studio>

Methods - Managed Instance link — <https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/managed-instance-link-feature-overview> - Log Replay Service — <https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/log-replay-service-migrate> - Native backup & restore (MI) — <https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/restore-sample-database-quickstart> - Transactional replication (SQL DB) — <https://learn.microsoft.com/en-us/azure/azure-sql/database/replication-to-sql-database> - Import/Export · BACPAC — <https://learn.microsoft.com/en-us/azure/azure-sql/database/database-import> - bcp utility — <https://learn.microsoft.com/en-us/sql/tools/bcp-utility> - Smart Bulk Copy — <https://learn.microsoft.com/en-us/samples/azure-samples/smartbulkcopy/smartbulk-copy/> - Backup to URL — <https://learn.microsoft.com/en-us/sql/relational-databases/backup-restore/sql-server-backup-to-url> - Availability group → Azure VM — <https://learn.microsoft.com/en-us/data-migration/sql-server/virtual-machines/availability-group-migrate> - Log shipping — <https://learn.microsoft.com/en-us/sql/database-engine/log-shipping/about-log-shipping-sql-server> - Elastic Jobs (SQL DB) — <https://learn.microsoft.com/en-us/azure/azure-sql/database/elastic-jobs-overview>

Targets — containers, Fabric, AVS, Arc - Create Arc-enabled SQL Managed Instance — <https://learn.microsoft.com/en-us/azure/azure-arc/data/create-sql-managed-instance> - Migrate to Arc-enabled SQL MI — <https://learn.microsoft.com/en-us/azure/azure-arc/data/migrate-to-arc-enabled-sql-managed-instance> - SQL Server on Kubernetes / AKS — <https://learn.microsoft.com/en-us/sql/linux/quickstart-sql-server-containers-kubernetes> - SQL Server in a Docker container — <https://learn.microsoft.com/en-us/sql/linux/quickstart-install-connect-docker> - Fabric Migration Assistant for SQL database (Preview) — <https://learn.microsoft.com/en-us/fabric/database/sql/migration-assistant> - Migrate to Fabric SQL DB via DACPAC — <https://learn.microsoft.com/en-us/fabric/database/sql/migrate-with-migration-assistant-using-dacpac> - Fabric Mirroring for SQL Server — <https://learn.microsoft.com/en-us/fabric/mirroring/sql-server> - Limitations for SQL database in Fabric — <https://learn.microsoft.com/en-us/fabric/database/sql/limitations> - Fabric — choose a data store (decision guide) — <https://learn.microsoft.com/en-us/fabric/fundamentals/decision-guide-data-store> - Azure VMware Solution — <https://learn.microsoft.com/en-us/azure/azure-vmware/introduction> - SQL MI — Next-gen General Purpose — <https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/service-tiers-next-gen-general-purpose-use> - Azure SQL DB Hyperscale — <https://learn.microsoft.com/en-us/azure/azure-sql/database/service-tier-hyperscale>

Data movement & VM-level - Azure Data Box / Data Box Heavy — <https://learn.microsoft.com/en-us/azure/databox/data-box-overview> - ExpressRoute — <https://learn.microsoft.com/en-us/azure/expressroute/expressroute-introduction> - AzCopy — <https://learn.microsoft.com/en-us/azure/storage/common/storage-use-azcopy-v10> - Azure Storage Mover — <https://learn.microsoft.com/en-us/azure/storage-mover/service-overview> - Azure Site Recovery — <https://learn.microsoft.com/en-us/azure/site-recovery/site-recovery-overview> - Azure-SSIS Integration Runtime — <https://learn.microsoft.com/en-us/azure/data-factory/create-azure-ssis-integration-runtime> - SQL Data Sync (retiring 30 Sep 2027) — <https://learn.microsoft.com/en-us/azure/azure-sql/database/sql-data-sync-data-sql-server-sql-database?view=azuresql>

Licensing / programs / lifecycle - Azure Hybrid Benefit (Azure SQL) — <https://learn.microsoft.com/en-us/azure/azure-sql/azure-hybrid-benefit> - SQL Server ESU enabled by Azure Arc — <https://learn.microsoft.com/en-us/sql/sql-server/azure-arc/extended-security-updates> - Move SQL Server license to PAYG (Arc) — <https://learn.microsoft.com/en-us/sql/sql-server/azure-arc/manage-pay-as-you-go-transition> - SQL Server end-of-support / ESU — <https://learn.microsoft.com/en-us/sql/sql-server/end-of-support/sql-server-extended-security-updates> - SQL Server 2016 lifecycle — <https://learn.microsoft.com/en-us/lifecycle/products/sql-server-2016> - Azure SQL MI free offer — <https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/free-offer> - Azure SQL DB free offer — <https://learn.microsoft.com/en-us/azure/azure-sql/database/free-offer> - Azure Accelerate (partner programs) — <https://partner.microsoft.com/en-us/partnership/azure-offerings> - Tool consolidation / retirement (blog) — <https://www.microsoft.com/en-us/sql-server/blog/2024/09/12/modernize-your-database-with-the-consolidation-and-retirement-of-azure-database-migration-tools/> - Cloud Adoption Framework — migrate — <https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/migrate/> - Azure landing zones (CAF) — <https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/landing-zone/> - Savings plan for databases — <https://learn.microsoft.com/en-us/azure/cost-management-billing/savings-plan/savings-plan-overview> - Modernize your databases (hub) — <https://aka.ms/modernizedatabases>

*Links last verified: July 2026. Microsoft migration guides moved from .../azure-sql/migration-guides/... to .../data-migration/sql-server/... (redirects in place). Items marked **Preview** are subject to change.*

17 Document version & changelog

Current version: **v1.1** (2026-07-03).

Version history (current: v1.1)

Version	Date	Changes
v1.1	2026-07-03	Azure SQL MI Next-gen General Purpose re-classified preview → GA (GA since Nov 2025). Dates refreshed to July 2026 (verification note, lifecycle status column, sources footer). All ~45 Microsoft Learn links re-verified. Re-confirmed still-current: Fabric Migration Assistant remains Preview (DACPAC ≤ 20 MB, on-prem gateway only, no Private Link); SSMS 22 Azure SQL assessment still roadmap ~Q3 CY2026; SQL Server 2016 ESU free on Azure from 14 Jul 2026; MI Link GA with SQL Server 2025 support + reverse fallback to 2022/2025.
v1.0	2026-06-06	Initial published knowledge base: 8 target families (SQL VM, AVS, SQL MI, SQL DB, Fabric SQL DB, containers AKS/ARO/ACI/ACA, Arc-enabled SQL MI, Arc in-place); targets/control-planes/methods taxonomy; per-target method tables (min source · downtime · constraints); 2025–2026 tooling reset (DMA, ADS, DMS classic, SQL Data Sync retirements); tooling-by-source matrix (incl. STRIIM online to Azure SQL DB); downtime strategy; summary & decision matrices; ancillary components; cross-cloud & reverse migration; third-party alternatives; field insights; commercial & funding levers (AHB / ESU / PAYG · Savings plan for databases · Azure Accelerate); FY27 SQL Motion context & AI Migration Agent I/O contract.

Versioning: **MAJOR.MINOR** — MINOR bumps for fact/link refresh and additive content; MAJOR bumps for structural rewrites or scope changes.

18 Appendix — Using the sql-migration-advisor Copilot CLI skill

This knowledge base also powers sql-migration-advisor, a GitHub Copilot CLI skill that turns the inventory above into a guided recommendation. Ask Copilot “*migrate a SQL Server environment to Azure*”: it runs a short ~10-question interview (scope, source location & version, primary driver, instance-level feature dependencies, largest DB size, downtime tolerance, network & ports, compliance, ancillary services), then returns a grounded recommendation card — target, method, downtime class, blockers + remediations, cost levers, and the right Microsoft program — and never recommends retired tooling (DMA, Azure Data Studio extension, DMS classic).

18.1 Install (once)

macOS / Linux:

```
git clone https://github.com/fredgis/sql-migration-advisor.git
↪ ~/.copilot/skills/sql-migration-advisor
```

Windows (PowerShell):

```
git clone https://github.com/fredgis/sql-migration-advisor.git
↪ "$env:USERPROFILE\.copilot\skills\sql-migration-advisor"
```

Restart Copilot CLI, run /skills to confirm sql-migration-advisor is listed, then ask your question in natural language (English or French).

18.2 Example output

The card below is a real recommendation for an estate of ~100 on-premises SQL Server 2012 databases (limited WAN, exiting out-of-support):

Verdict - estate of ~100 databases (on-prem SQL 2012 + PaaS)
Azure SQL Managed Instance (General Purpose) via LRS in seed-then-sync · downtime minimal · grounded in the FY27 SQL knowledge base.

Fully-managed PaaS + “not sure” dependencies + exiting out-of-support SQL 2012: SQL MI is the default managed lift-and-shift. It absorbs your 150 GB+ TB databases in General Purpose, handles any undocumented dependencies (linked servers, SQL Agent), and MI Link is excluded because the 2012 source is below the required 2016+ → the method is LRS.

Estate strategy (100 databases)

1. Azure Migrate: agentless discovery, dependency mapping, costed business case, and MI right-sizing.
2. **Az.DataMigration** (CLI/PowerShell): industrializes LRS beyond ~50 databases - the only viable path at this scale.
3. Data Box for the initial seed (limited WAN), then LRS catches up the log delta before each cutover.

At a glance

	Recommendation
Target	Azure SQL Managed Instance - General Purpose (size 150 GB+ TB, managed PaaS)
Method	Log Replay Service (LRS) - 2012 source OK; MI Link excluded (requires 2016+)
Downtime	Minimal (LRS replays full/diff/log, short cutover)
Assess / orchestrate	Azure Migrate → Az.DataMigration at scale → SSMS 22 / Arc

Blockers & fixes

- Limited WAN + multi-TB + Data Box for the initial seed, then LRS sync (seed-then-sync pattern); otherwise timeouts are guaranteed.
- Out-of-support SQL 2012 source → migrate fast; restoring via LRS removes the need for ESU.
- “Not sure” dependencies → run an Azure Migrate dependency map before locking the target <40% of migrations stumble on forgotten linked servers / SQL Agent jobs).
- Logins / users → script and recreate (not migrated by default).
- EU data residency → MI region within the EU perimeter from creation.
- Network → validate outbound nls (9140) and nls across all 100 sources.

Cost & program (your question)

- Azure Hybrid Benefit (AHB) - eligible on SQL MI (Cores) - reuse your SQL licenses with SA, up to ~80% savings combined with 1-3 year reservations.
- ESU = the real “stay vs migrate” lever: staying on SQL 2012 on-prem forces paid ESU via Azure Arc; migrating to MI (evergreen) eliminates that cost entirely.
- Consolidation: 100 databases → group onto a few MI instances (instance pools / shared vCores) instead of 100 isolated targets - a major cost lever.
- GP vs RC: stay General Purpose (pay-per-minute storage) unless you need local latency + Business Critical.
- Sizing: baseline Perform ≥ 7 days + ~20% headroom, never average CPU - over-sizing across 100 databases is expensive.
- Programs: Cloud Accelerate Factory (zero-cost delivery), SQL in a Day, Azure Accelerate / FastTrack - attach to fund the delivery.

⚠ Biggest risk: network throughput over a limited WAN with 100 databases. Don't compute **size + bandwidth** - test AzCopy + one real full backup, seed via Data Box, and plan a rollback window. Validate target performance with DEA (capture + replay) on sensitive databases.

🔗 LRS · Az.DataMigration · Azure Migrate - business case · AHB - LRS: planned offline, no read-only target.

Would you like a per-database-group table, a scaled LRS runbook, or a summary slide for the estate?